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Type 2 diabetes care: knowledge and clinical practice among medical practitioners at Medical Teaching Institute Mardan, Pakistan – a cross-sectional study

Muhammad Maaz¹ , Muhammad Yasir¹ , Anosha Siddique¹ , Bilal Qammar² and Noorul Hadi^{1*}

Abstract

Background Type 2 Diabetes Mellitus (T2DM) poses a major health burden in Pakistan, where a shortage of endocrinologists places the responsibility of care on medical practitioners. Global studies reveal significant gaps in GP knowledge and practices. However, no such data exists for Pakistan, particularly in the Mardan region.

Methods This analytical cross-sectional study was conducted at Mardan Medical Complex, Pakistan, between January and March 2025, targeting licensed medical practitioners who had completed their MBBS and one-year internship. Participants were recruited using convenience sampling, and data were collected through a 34-item questionnaire covering demographics, T2DM knowledge, practices, barriers, confidence, and attitudes. Statistical analyses were performed in SPSS (IBM v27.0), using descriptive statistics and logistic regression, with a significance level of $p < 0.05$.

Results A total of 178 medical practitioners participated. The mean diabetes knowledge score was 9.05 ± 2.52 out of 14. Significant predictors of higher knowledge included managing >40 diabetic patients per week ($p = 0.002$), and working in medical rather than surgical specialties ($p = 0.007$). Years of practice was also a significant predictor ($p = 0.043$). Only 52.8% correctly identified the diagnostic HbA1c threshold ($\geq 6.5\%$), 64.6% the fasting glucose cut-off (≥ 126 mg/dL), 74.72% the random glucose level (≥ 200 mg/dL). While Caring for Diabetes patients 97.8% advised lifestyle modification and 91% utilized HbA1c testing, only 51.1% routinely performed foot exams. Major reported barriers included high treatment costs (84.8%), poor patient compliance (81.5%), limited specialist access (52.8%), and absence of clinical protocols (50.6%). Importantly, 93.3% of practitioners expressed willingness to attend diabetes-focused CME programs.

Conclusion Medical practitioners at public hospitals show moderate knowledge of type 2 diabetes, with good awareness of symptoms but notable gaps in diagnosis, guideline adherence, and complication screening.

Keywords Diabetes mellitus type 2, T2DM, Knowledge, Practice, Primary care, Medical practitioners

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Introduction

Diabetes Mellitus, also known as DM, is a common chronic metabolic syndrome marked by elevated blood sugar levels and impaired carbohydrate metabolism as a result of decreased insulin secretion or action. Mainly diabetes is of 2 types: type 1, which results from the destruction of pancreatic beta cells, and type 2 diabetes which is caused by the non-responsiveness of the body to insulin's action [1].

Diabetes is one of the most prevalent diseases globally, affecting approximately 536.6 million people, with type 2 diabetes contributing to approximately 90% of the cases [2]. According to a report of 2022 by the Institute for Health Metrics and Evaluation, diabetes accounts for around 1.6 million deaths annually, with the majority linked to type 2 diabetes [3]. Since diabetes is particularly prevalent in middle and low-income countries placing Pakistan at a greater risk of diabetes related-mortalities. According to the 2022 reports by the International Diabetes Federation 26.7% prevalence of diabetes among adults was recorded bringing the total number of cases to approximately 33 million in Pakistan [4], this number is very alarming and increasing with each year.

Although diabetes is not directly fatal, it is associated with severe complications such as heart failure, chronic kidney disease, and other conditions that significantly increase mortality risk among the type 2 diabetes population [5]. Clinical trials such as the “Diabetes Control and Complications Trial” (DCCT) and the “Uk Prospective Diabetes Study” (UKPDS) indicate that controlling blood glucose level, cholesterol and blood pressure significantly reduces serious complications of type 2 diabetes such as cardiovascular, renal & diabetic neuropathy [6, 7]. However, according to “Second National Diabetes Survey Pakistan” glycaemic dysregulation was found to be 43.8% and 39.3% in urban and rural areas of Pakistan respectively, also risk factors for diabetes such as hypertension, obesity and dyslipidaemia were recorded to be significantly high [8].

Diabetes being a chronic condition requires intense care and diabetic specialists or endocrinologists are arguably the most competent of all due to their specialized training but regrettably, most developing countries are facing acute shortage of endocrinologists. For instance, Pakistan has one endocrinologist for every 200,000 patients [9]. Due to such shortage medical practitioners constitute the major diabetic care manpower in most developing countries.

Several studies have been conducted in many countries on evaluation of medical Practitioners' knowledge and clinical practice. A study in Iran showed that the general practitioner's knowledge and clinical practice were not satisfactory, showing a very significant gap between standard of patient care and current patients care. New

strategies and training are strongly emphasized for better patients care [10].

A study in south eastern Nigeria exposed serious gaps in knowledge and practices in diabetes care among primary care physicians, with the majority lacking the knowledge of proper diagnostic criteria and showing a lack of interest and participation in continuing medical education (CME) programs [11].

Another study in Shiraz, Iran on diabetes mellitus management also highlighted significant shortcoming in the diagnostics and treatment of the disease [12].

Other studies evaluating medical practitioners' knowledge and clinical practice in China [13], Central and Eastern Europe [14], India [15], Australia [16] and a survey of the Point-Of-Care Information Technology (POC-IT) database [17] indicated similar unsatisfactory results and emphasized on proper training of general physicians through modern CME programs.

In low- and middle-income countries, diabetes is most often managed by medical practitioners from various specialties due to limited resources and a shortage of endocrinologists. Global studies have highlighted significant gaps in their knowledge and practice, contributing to suboptimal diabetes care and increased risk of complications. This highlights the importance of evaluating the knowledge and practices of medical practitioners in Mardan, Pakistan, to identify gaps and inform strategies for improved diabetes care.

Methods

Study design and sampling technique

A cross-sectional study was conducted at Mardan Medical Complex, a major tertiary care hospital in Mardan, Pakistan, offering comprehensive healthcare services and serving as a training centre for medical professionals. The study was conducted between January 2025 and March 2025, after obtaining ethical approval from the Research and Ethics Committee of Bacha Khan Medical College. Participants were recruited using a convenience sampling technique and included those who were available and provided informed consent.

Participants

This study targeted medical practitioners who had completed their formal medical school (MBBS) along with a one-year internship.

Sample size calculations were performed using OpenEpi Version 3, open-source calculator (SSPropor) based on a finite population of 378 doctors working at Mardan Medical Complex. Sample Size Calculations were based on a margin of error of 0.05, a confidence interval of 95%, an average knowledge score of 62.5% among family doctors (Haghighinejad et al.) [12]. A sample size of 185 participants was calculated using these parameters. During

the study period, despite efforts to recruit the intended number of participants, the response rate was lower than anticipated due to limited engagement from practitioners, with several declining to provide consent or complete the questionnaire despite multiple attempts.

Inclusion criteria

- Licensed Medical Practitioners currently employed at Mardan Medical Complex.
- Doctors who have a MBBS degree who have completed their Internship (House job).

Exclusion criteria

- Doctors doing their internship (House Job).
- Doctors who declined to participate or submitted incomplete questionnaires.

Objectives

The primary objective of this study was to assess the knowledge and clinical practices of medical practitioners regarding type 2 diabetes management. The secondary objectives included identifying barriers faced in providing optimal diabetes care and exploring practitioners' attitudes toward continuing education and training in diabetes management.

Data collection tool

Data were collected using a structured, self-administered questionnaire in English, adapted with permission from Ugwa et al. (2020) [18]. To ensure validity, the adapted questionnaire was reviewed for content relevance, clarity, and contextual appropriateness by subject experts prior to data collection. Minor modifications were made to the knowledge section, in which open-ended questions were converted into closed-ended items to improve standardization and ease of analysis, while preserving the original constructs. A pilot test was conducted on a subset of participants to assess clarity and feasibility, and no further changes were required. The internal consistency reliability of the questionnaire was assessed using Cronbach's alpha, which yielded a value of 0.701, indicating acceptable reliability.

Demographics

This section collected background data including age, sex, years of clinical practice, medical or surgical specialty, professional designation, type (government/private) and location (urban/rural) of practice, weekly diabetic patient load, access to HbA1c testing facilities, and prior participation in diabetes-related workshops. This information helped contextualize the professional profile and practice setting of each respondent.

Knowledge about diabetes

This section assessed the fundamental knowledge related to knowledge of diabetes, including classical symptoms, modifiable risk factors, and diagnostic thresholds for fasting blood glucose (FBS), Random blood glucose (RBS) and HbA1c for Diabetes and Pre-diabetes. It also assessed awareness of existing diabetes management guidelines.

Diabetes care practice

This portion examined the actual clinical behaviour of doctors in managing diabetes. It inquired about the diagnostic tests routinely used, physical and laboratory examinations conducted for new diabetic patients, counselling areas (e.g., lifestyle, adherence), antidiabetic medications commonly prescribed, and the parameters regularly monitored for glycaemic control and complication screening. Referral practices to endocrinologists were also explored.

Barriers to diabetes care, confidence in managing diabetes, and attitude toward workshops

This section aimed to identify the key obstacles medical practitioners face in providing optimal diabetes care. Participants were asked to select from a list of common challenges such as limited access to diagnostic tools, lack of diabetes specialists, cultural influences, excessive workload, and poor patient compliance.

Additionally, the questionnaire assessed the practitioners self-reported confidence in various aspects of diabetes management, including selecting appropriate oral medications, initiating insulin therapy, and managing diabetes overall, using a 3-point Likert scale, with 1 indicating low confidence, 2 indicating moderate confidence, and 3 indicating high confidence.

Finally, the section explored the participants' attitudes toward continuing education by asking whether they were willing to attend periodic training workshops on diabetes management.

Statistical plan

Data was extracted from the questionnaires and analysed with the Statistical Package for Social Sciences (SPSS) (IBM version 27.0; SPSS). Only the data from fully completed questionnaires was eligible for data analysis. Categorical variables were presented as numbers and percentages while continuous variables were presented as means and standard deviations.

A cumulative Knowledge Score (maximum 14 points) was calculated by awarding 1 point for each correct response and 0 for incorrect or blank answers. Three points were allotted each for correctly listing classical symptoms and modifiable risk factors (1 point per correct option). One point was given for each correct

diagnostic threshold (fasting glucose, random glucose, HbA1c for both diabetes and prediabetes). Additionally, participants received 1 point per correctly named professional guideline, with a maximum of two points. This score was treated as a continuous variable to assess overall knowledge distribution and as a categorical variable, with scores dichotomized into high knowledge (10–14 points) and low-to-moderate knowledge (0–9 points).

To identify predictors of higher diabetes knowledge, binary logistic regression was performed. Initially, univariable logistic regression was conducted to explore associations between each independent variable and knowledge score. Variables with p -values <0.25 in univariable analysis were included in a multivariable logistic regression model to control for confounding. Odds ratios (OR) and adjusted odds ratios (aOR) with 95% confidence intervals (CI) were reported. A p -value <0.05 was considered statistically significant.

Table 1 Detailed profile of studied practitioners

Variable	Frequency	Percentages (%)
Age		
Mean $\pm$ SD	29.15 $\pm$ 7.76	
Sex		
Male	113	63.48%
Female	65	36.52%
Practice Years		
Median (IQR)	3 (2 – 4)	
Practice Field		
Medicine	79	44.38%
Surgery	99	55.62%
Designation		
Training Resident	152	85.39%
Assistant Professor	18	10.11%
Associate Professor	4	2.25%
Professor	1	0.56%
Medical Officer	3	1.69%
Practice Location		
Urban	171	96.07%
Rural	7	3.93%
Patients per Week		
≤ 10	48	26.97%
11–20	51	28.65%
21–30	39	21.91%
31–40	15	8.43%
>40	25	14.04%
HbA1c Facility		
No	4	2.25%
Yes	174	97.75%
Last Diabetes Workshop Attended		
Never	53	29.78%
Yes	125	70.22%
1 to 2 Years	105	84%
3 to ≥ 4 Years	15	12%
≥ 5 Years	5	4%

Data regarding clinical practices (e.g., diagnostic tests, examinations, lab assessments, counselling topics, medication use, complication screening, and referral practices) were analysed using frequencies and percentages. Barriers to care, self-perceived competence, and willingness for training were similarly described using descriptive statistics.

Results

Demographics

Table 1 shows that a total of 178 medical practitioners participated in the study, achieving an 96.2% response rate, with a mean age of 29.1 years and average clinical experience of 4.5 years. Most were male (63.5%), training residents (85.4%), and worked in public-sector hospitals. A majority (97.8%) reported access to HbA1c testing, while 59.0% had attended a diabetes-focused workshop within the past 1–2 years. Patient load varied, with 28.7% seeing 11–20 type 2 diabetes patients weekly and 14.0% seeing more than 40. When asked about prior exposure to diabetes-specific continuing medical education, 59.0% had attended a workshop within the past 1–2 years, whereas 29.8% had never attended one.

Knowledge about type 2 diabetes mellitus

Table 2 reveals that overall, the assessment of medical practitioners' knowledge regarding type 2 diabetes mellitus (T2DM) showed moderate understanding with notable gaps. Most respondents correctly identified classic symptoms such as polyuria (97.2%), polydipsia (93.3%), and polyphagia (83.7%), and recognized obesity (93.8%) and sedentary lifestyle (92.1%) as modifiable risk factors. However, fewer correctly identified diagnostic cut-offs for prediabetes and diabetes, with only 56.2% selecting the correct fasting blood sugar (FBS) (100–125 mg/dL), and 52.8% choosing the correct HbA1c ($\geq 6.5\%$) threshold. Regarding professional guidelines, 71.9% reported not being aware of any, while only 28.1% could name at least one valid source. A cumulative knowledge scores out of 14 was calculated based on correct responses, with scores ranging from 2 to 14. Among the participants, 9 (5.1%) had low knowledge, 85 (47.8%) had moderate knowledge, and 83 (46.6%) had high knowledge, indicating that most practitioners had moderate to high knowledge regarding type 2 diabetes mellitus. Figure 1 depicts the distribution of cumulative knowledge scores (1–14) with the percentage of participants in each category.

Predictors of knowledge among medical practitioners

Logistic regression analysis was performed to identify predictors of higher diabetes-related knowledge scores among medical practitioners. In univariable analysis, significant associations included practice years (OR: 1.062; 95% CI: 1.002–1.126; $p = 0.043$), surgical practice field

Table 2 Summary of diabetes knowledge responses and overall knowledge scores

Variable	Frequency	Percentages (%)
Knowledge Score		
Low	9	5.1%
Moderate	85	47.8%
High	83	46.6%
List 3 Classical Symptoms of Diabetes		
Polyuria	173	97.19%
Weight Gain	25	14.04%
Polydipsia	166	93.26%
Polyphagia	149	83.71%
Nausea	10	5.62%
List 3 Modifiable Risk Factors for Diabetes		
Sedentary lifestyle	164	92.13%
Obesity	167	93.82%
Family History	56	31.46%
Age	16	8.99%
High Blood Pressure	112	62.92%
FBS for Prediabetes		
200-250	0	0%
100-125	100	56.18%
115-130	28	15.73%
126-150	37	20.79%
90-120	13	7.30%
RBS for Prediabetes		
100-126	7	3.93%
120-135	15	8.43%
126-199	32	17.98%
140-199	103	57.87%
200-249	21	11.80%
HbA1c for Prediabetes		
5.2-6.2	5	2.81%
5.4-6.0	15	8.43%
5.7-6.4	41	23.03%
6.0-7.0	59	33.15%
6.5-8.0	58	32.58%
FBS for Diabetes		
≥100	6	3.37%
≥120	30	16.85%
≥126	131	73.60%
≥200	8	4.49%
≥95	3	1.69%
RBS for Diabetes		
≥126	7	3.93%
≥135	1	0.56%
≥140	30	16.85%
≥200	133	74.72%
≥250	7	3.93%
HbA1c for Diabetes		
≥6.0	23	12.92%
≥6.5	94	52.81%
≥7.0	23	12.92%
≥7.5	33	18.54%
≥8.0	5	2.81%

Table 2 (continued)

Variable	Frequency	Percentages (%)
Do You Know Any Professional Guidelines for Managing DM?		
No	128	71.91%
Yes	50	28.09%

Abbreviations: FBS Fasting Blood Sugar, RBS Random Blood Sugar, HbA1c Glycated Haemoglobin, DM Diabetes Mellitus

(OR: 0.435; 95% CI: 0.237–0.797; $p = 0.007$), and higher patient load per week, particularly those seeing 11–20 patients (OR: 3.35; 95% CI: 1.45–7.76; $p = 0.005$) and >40 patients (OR: 5.16; 95% CI: 1.81–14.68; $p = 0.002$). In the multivariable model, surgical specialty remained a significant negative predictor of knowledge (aOR: 0.479; 95% CI: 0.231–0.993; $p = 0.048$), indicating lower knowledge scores compared to practitioners in medicine. Additionally, those seeing 11–20 patients (aOR: 3.40; 95% CI: 1.37–8.42; $p = 0.008$) and >40 patients per week (aOR: 3.59; 95% CI: 1.11–11.57; $p = 0.032$) were significantly more likely to have higher knowledge scores. While practice years approached significance in the adjusted model (aOR: 1.056; 95% CI: 0.99–1.13; $p = 0.099$), sex, designation, and prior workshop attendance did not show significant associations in either model Table 3.

Diabetes care & practice

Table 4 presents the diagnostic approaches, highlighting the most commonly used methods were HbA1c (91.0%), fasting blood sugar (79.2%), and random blood sugar (65.2%), while fewer practitioners reported using oral glucose tolerance tests (27.5%) and urine glucose (13.5%).

Routine examinations performed during follow-up varied, with blood pressure monitoring (70.8%) and foot exams (61.2%) being more common than BMI measurement (60.1%), eye exams (47.8%), and waist circumference (19.7%). Similarly, routine laboratory assessments included HbA1c (96.6%), urea/creatinine (64.6%), lipid profile (55.1%), and urine analysis (48.9%), while ECGs were reported by 32.0%.

Most participants reported consistently counselling patients on lifestyle modification (97.8%), medication adherence (77.0%), self-monitoring (75.8%), and hypoglycemia prevention (64.6%). Counselling on lipid management (43.3%) and foot care (68.5%) was also practiced, though less consistently.

In terms of pharmacologic management, the most frequently prescribed drug classes were sulfonylureas (83.7%), insulin (82.6%), biguanides (74.2%), and multivitamins (57.9%). Newer agents like DPP-4 inhibitors (44.9%) and thiazolidinediones (29.8%) were used less often, and alpha-glucosidase inhibitors (21.4%) were the least commonly prescribed.

Monitoring practices showed that most practitioners regularly tracked fasting blood glucose (93.8%),

Knowledge Score Distribution

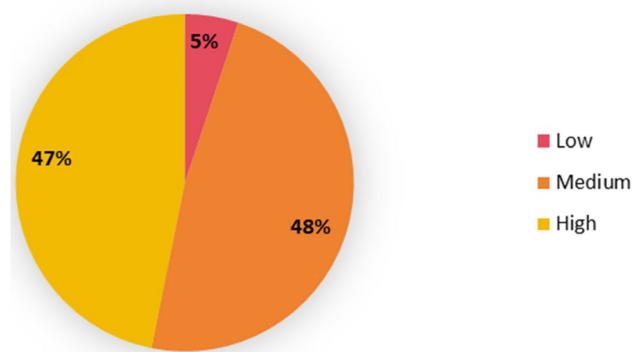


Fig. 1 Distribution of knowledge scores among participants

postprandial glucose (61.2%), and HbA1c (89.9%). Weight monitoring was comparatively lower (39.9%).

For complication screening, annual assessments were most commonly done for renal function (72.5%), eye health (67.4%), and urine microalbumin (68.5%), followed by lipid profile (58.4%) and foot examinations (51.1%).

Referral to an endocrinologist was typically made in cases of poor glucose control (75.3%) or when complications were detected (66.3%). However, 12.4% reported never referring patients to a specialist under any circumstances (Table 4).

Barriers in diabetes care confronted by gps

Commonly reported barriers to effective diabetes care included affordability (84.8%), poor patient adherence (81.5%), lack of specialists (52.8%), allied care shortages (59.0%), and absence of clinical protocols (50.6%). Equipment limitations and workload were also cited by 38.8% each, and 23.0% mentioned cultural or religious issues (Figure 2).

Self-perceived competence and training willingness

Figure 3 shows that, although 72.5% reported routinely screening asymptomatic adults for diabetes, a substantial 33.1% did not. In terms of self-assessment, most rated their knowledge of oral antidiabetic drug selection as moderate (57.3%), while 23.0% rated it low and 19.7% high. Confidence in initiating insulin was similarly mixed: 45.5% reported moderate confidence, 29.8% low, and 24.7% high. For overall diabetes management, 55.6% felt their skills were intermediate, 25.8% rated them as basic, and only 18.5% considered them advanced. Importantly, 93.3% of participants expressed willingness to receive periodic training in diabetes management.

Table 3 Predictors of knowledge among medical practitioners

Predictor	Univariable Analysis		Multivariable Analysis	
	OR (Lower - Upper)	p-value	OR (Lower - Upper)	p-value
Age	1.03 (0.99–1.07)	0.214	/	
Practice Years	1.06 (1.00–1.13)	0.043	1.06 (0.99–1.13)	0.099
Sex				
Male	1 - Reference			
Female	0.67 (0.36–1.25)	0.21	1.25 (0.58–2.66)	0.568
Practice Field				
Medicine	1 - Reference			
Surgery	0.44 (0.24–0.80)	0.007	0.48 (0.23–0.99)	0.048
Designation				
Medical Officer	1 - Reference		/	
Training Medical Officer	1.68 (0.15–18.96)	0.674	/	
Assistant Professor	4.00 (0.30–53.47)	0.295	/	
Associate Professor	0.67 (0.03–18.06)	0.81	/	
Professor		1	/	
Diabetes Workshop				
Never	1 - Reference			
1–2 Years	1.57 (0.80–3.09)	0.192	1.21 (0.57–2.59)	0.625
3–4 Years	1.83 (0.57–5.82)	0.307	1.74 (0.50–6.04)	0.385
≥5 Years	2.40 (0.37–15.64)	0.36	3.90 (0.53–28.44)	0.18
Patients Per Week				
≤10	1 - Reference			
11 to 20	3.35 (1.45–7.76)	0.005	3.40 (1.37–8.42)	0.008
21 to 30	1.88 (0.77–4.56)	0.165	1.71 (0.62–4.68)	0.297
31 to 40	1.62 (0.49–5.41)	0.434	1.41 (0.39–5.07)	0.598
>40	5.16 (1.81–14.68)	0.002	3.59 (1.11–11.57)	0.032

Discussion

This study provides a comprehensive assessment of the knowledge, clinical practices, and perceived barriers among medical practitioners managing type 2 diabetes mellitus (T2DM) in a public-sector hospital in Pakistan. The results of the current research do not fully agree with previously published literature on several key points, demonstrating clear contextual contrasts rather than mere repetition of global trends. In line with our overall finding of a moderate knowledge level (mean score 9.05 out of 14), our respondents demonstrated average general knowledge with particularly significant deficiencies

Table 4 Summary of diabetes care practices among medical practitioners

Variable	Frequency	Per-cent-ages (%)
Diagnostic tests used for diabetes		
Fasting Blood Sugar	141	79.21%
Random Blood Sugar	116	65.17%
HbA1c	162	91.01%
OGTT	49	27.53%
Urine Glucose	24	13.48%
Clinical exams performed in newly diagnosed T2DM		
Blood Pressure	126	70.79%
BMI	107	60.11%
Waist Circumference	35	19.66%
Foot Exam	109	61.24%
Eye Exam	85	47.75%
Laboratory tests ordered for new T2DM cases		
Urine Analysis	87	48.88%
HbA1c	172	96.63%
Lipid Profile	98	55.06%
Urea/Creatinine	115	64.61%
ECG	57	32.02%
Routine diabetes counselling topics provided		
Lifestyle Modification	174	97.75%
Medication Adherence	137	76.97%
Hypoglycemia	115	64.61%
Lipid Profile	77	43.26%
Self-Monitoring	135	75.84%
Foot Care	122	68.54%
Classes of anti-diabetic medications prescribed		
Sulphonylureas	149	83.71%
Biguanides	132	74.16%
Thiazolidinediones	53	29.78%
Alpha-glucosidase inhibitors	38	21.35%
DPP4 Inhibitors	80	44.94%
Insulin	147	82.58%
Multivitamins	103	57.87%
Glycemic/metabolic parameters regularly monitored		
FBS	167	93.82%
Post meal blood glucose	109	61.24%
HbA1c	160	89.89%
Weight	71	39.89%
Routine screening for diabetes complications		
Annual dilated eye examination	120	67.42%
Annual urine micro albumin	122	68.54%
Annual comprehensive foot examination	91	51.12%
Annual Lipid profile	104	58.43%
Annual serum electrolyte urea and creatinine	129	72.47%
Indications for referral to endocrinologists		
Poor Glucose Control	134	75.28%
As soon as chronic complications are noticed	118	66.29%
Advanced Complications	116	65.17%
I do not refer patients to specialist for any reason	22	12.36%

in understanding diagnostic thresholds, including fasting plasma glucose and HbA1c cutoffs. However, the magnitude of this knowledge gap was greater than that reported by Ugwa et al. (2020), who observed relatively higher awareness of HbA1c levels among non-specialist doctors in Nigeria [11]. Similarly, regional studies from Lahore and Karachi have documented poor awareness of updated diabetes management guidelines, often attributed to limited exposure to structured continuing medical education (CME); nonetheless, the extent of diagnostic unawareness in our cohort suggests a wider disconnect between guideline dissemination and real-world practice at the frontline level in public-sector hospitals [18].

With regard to clinical practice, our findings revealed inconsistent use of neuropathy screening and routine foot examinations, in contrast to studies from India where such practices are more consistently implemented due to better availability of allied health support. A notable finding unique to our study, and not prominently reported in prior South Asian literature, was the positive correlation between higher patient load and improved practitioner knowledge scores. This observation may reflect the role of intensive clinical exposure in reinforcing experiential learning in the management of T2DM, particularly in overcrowded healthcare settings with limited access to organized CME programs, and is consistent with findings reported by Zhong et al. (2018) [19]. Conversely, medical practitioners from surgical specialties scored significantly lower, likely due to limited routine engagement with metabolic disorders during training a trend also reflected in literature from comparable healthcare systems [20]. Although longer years of practice showed a trend toward improved knowledge, this association did not remain significant after multivariable analysis, indicating that experience alone is insufficient without ongoing professional development [21].

In terms of clinical practice, the present study revealed encouraging adherence to recommended diagnostic approaches, including the use of fasting blood glucose and HbA1c for both diagnosis and monitoring. Participants also reported strong counselling practices concerning lifestyle modification and medication adherence. However, critical aspects of diabetes care such as lipid profiling, neuropathy screening, and preventive foot examinations were practiced less consistently. This contrasts with patterns observed in higher-resource health systems, where routine foot care and comprehensive metabolic screening are well integrated into primary care [22]. The discrepancy likely reflects contextual constraints in Pakistan, including limited availability of allied health professionals, high patient volumes in outpatient departments, resource shortages, and fragmented referral pathways. These systemic challenges may contribute

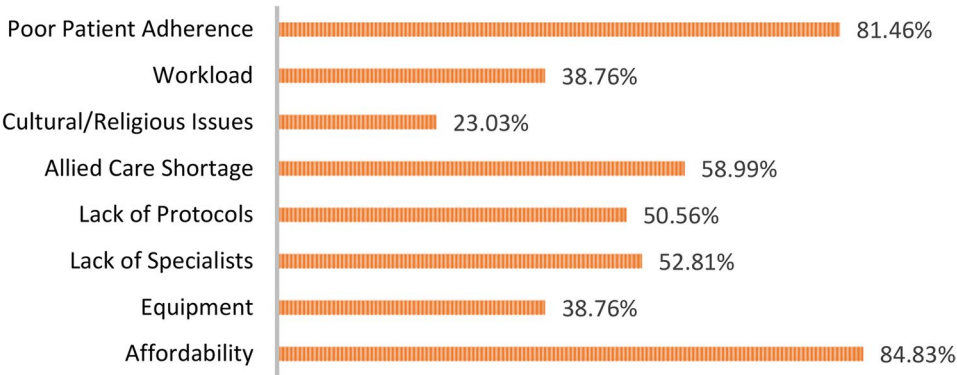


Fig. 2 Challenges faced by medical practitioners in managing diabetes care

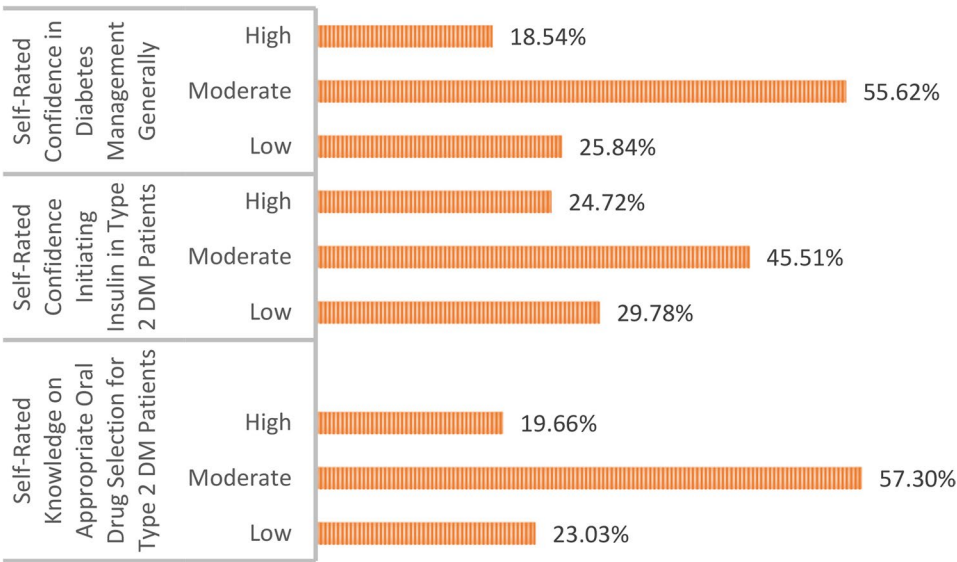


Fig. 3 Self-rated knowledge, clinical confidence, and willingness for diabetes training among medical practitioners

to the gap between medical practitioners knowledge and actual implementation of comprehensive diabetes care.

Pharmacologic treatment patterns also offer insight into contextual realities. Medical practitioners in the present study predominantly prescribed older, affordable medications such as metformin and sulfonylureas, with limited use of newer agents. Similar trends are reported in South Asian countries such as India and Bangladesh, where economic barriers, limited insurance coverage, and lack of familiarity with advanced therapies impede their uptake. Moreover, most participants rated their own competence as average or basic, a perception that aligned with their modest knowledge scores [23]. This finding underscores the need for structured CME programs and updated training curricula in diabetes care.

Several barriers to optimal diabetes care were identified in the present study, with affordability of treatment, poor patient adherence, and limited availability of diagnostic facilities being the most commonly cited. These challenges echo reports from the International Diabetes

Federation, highlighting shared obstacles across LMICs. Cultural and religious beliefs, insufficient allied health support, and lack of structured clinical protocols were also identified as important barriers [24]. Understanding these contextual barriers is crucial for interpreting why knowledge gaps persist despite physicians willingness for improvement. In contrast, higher-income countries such as Saudi Arabia and South Africa have demonstrated greater CME participation and stronger adherence to guidelines due to national diabetes programs—suggesting that policy-level reforms can substantially enhance primary care performance.

Beyond clinical care, this study’s findings align with international discussions about evolving diabetes classification systems incorporating genetic, pathophysiological, and phenotype-based clusters [25]. While such advanced stratification is not yet integrated into public-sector care in Pakistan, global developments underscore the need for countries to strengthen foundational clinical competence so future innovations can be effectively adopted. Overall,

the findings of this study emphasize the need for systemic interventions, including dissemination of national guidelines, capacity-building through CME, integration of allied health professionals, and context-sensitive patient education strategies. Strengthening these elements could substantially improve diabetes outcomes not only in Pakistan but also in other LMICs with similar health system constraints [26].

Limitations

This study has several limitations that should be acknowledged. First, its single-center design in a public-sector tertiary hospital restricts generalizability to private, rural, or primary care settings where resource availability and training may differ significantly, thereby limiting external validity. Second, the use of a self-administered and self-reported questionnaire introduces potential response and social desirability biases, as participants may have overestimated their knowledge or reported ideal rather than actual clinical practices. Third, voluntary participation raises the possibility of self-selection bias, where practitioners more interested in diabetes management may have been more likely to participate. Fourth, the cross-sectional nature of the study prevents causal inferences regarding the relationships between participants characteristics and knowledge or practice patterns. Finally, the study did not verify reported practices through direct observation or patient chart review, which limits the ability to validate clinical behaviour objectively.

Conclusion

It is concluded that medical practitioners at public sector hospitals demonstrate moderate knowledge and varied practices in the management of type 2 diabetes mellitus. While awareness of classical symptoms and common risk factors is relatively high, substantial gaps remain in the application of diagnostic criteria, adherence to clinical guidelines, and routine screening for complications. Clinical experience, particularly higher patient load, was positively associated with better knowledge, whereas surgical specialties lagged in familiarity with diabetes care protocols. To address these gaps, structured CME programs, improved dissemination of national guidelines, and development of local clinical protocols are recommended. Strengthening multidisciplinary diabetes care models may further support better patient outcomes.

Abbreviations

T2DM	Type 2 Diabetes Mellitus
DM	Diabetes Mellitus
CME	Continuing Medical Education
RBS	Random Blood Sugar
FBS	Fasting Blood Sugar
HbA1c	Glycated Hemoglobin
OGTT	Oral Glucose Tolerance Test
aOR	Adjusted Odds Ratio

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

MM conceived and designed the study, performed the literature search and analysis, and contributed to manuscript writing. MY contributed to the conception and design of the work and was responsible for data acquisition. AS contributed to data acquisition and manuscript writing. BQ contributed to manuscript writing. **NH** provided supervision and made substantial critical revisions to the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the principles of the Declaration of Helsinki. Ethical approval for this cross-sectional study was obtained from the Institutional Ethical Review Committee of Bacha Khan Medical College (Reference No. 626/BKMC). Written informed consent was obtained from all participants prior to enrolment.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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